Quantum Physics, 2026 Fall, BME
Lecture 04 - The Quantum Harmonic Oscillator

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1 Background: The classical harmonic oscillator

1. The harmonic oscillator is a minimal model of many physical systems: Mass on a spring. Normal vibrational modes of ball-and-spring systems, solid bodies, molecules, crystal lattices. Motion of a particle trapped in a local potential minimum. Electromagnetic modes and radiation.

2. Example: mass on a spring.

3. Energy, expressed with spring constant:
$$E(p,x) = \frac{p^2}{2m} + \frac{1}{2}kx^2 \tag{1}$$

4. Energy, expressed with eigenfrequency:
$$E(p,x) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2, \tag{2}$$
where $\omega = \sqrt{k/m}$.

5. Motion without driving, see Figure 1(a):
$$x(t) = x_{\max}\cos(\omega t - \varphi), \tag{3}$$
$$p(t) = -p_{\max}\sin(\omega t - \varphi), \tag{4}$$
$$x_{\max} = \sqrt{\frac{2E}{m\omega^2}}, \tag{5}$$
$$p_{\max} = \sqrt{2mE}. \tag{6}$$

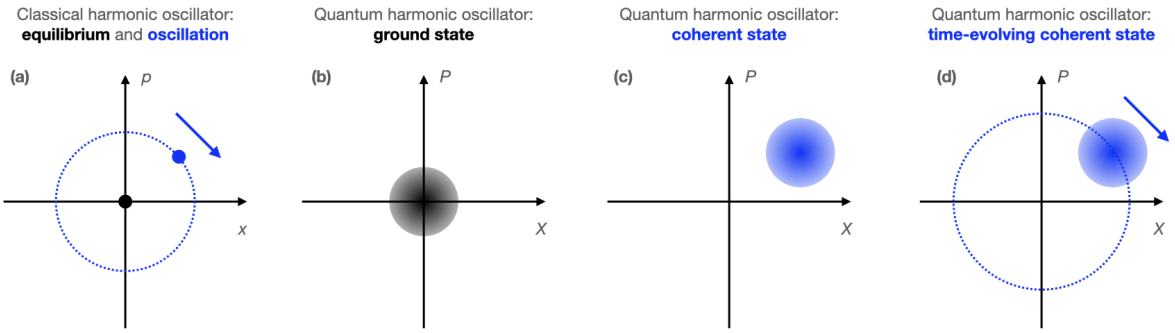


Figure 1: Dynamics of a classical harmonic oscillator, compared to a quantum harmonic oscillator's ground state, coherent state, and the dynamics of a coherent state.

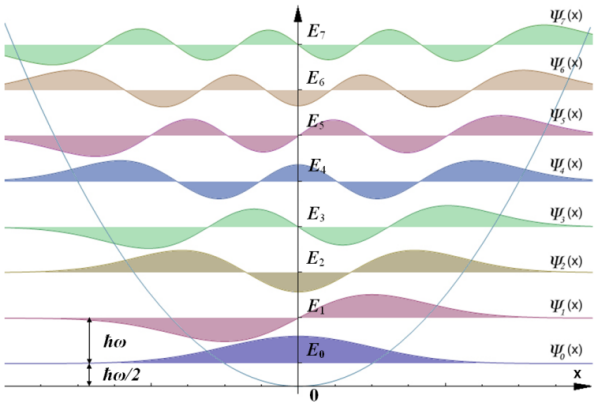


Figure 2: Eigenvalues and eigenfunctions/eigenvectors of the quantum harmonic oscillator with mass m and frequency ω. The characteristic spatial extension of the ground state ψ_0 is $\ell = \sqrt{\frac{\hbar}{m\omega}}$. Source: Wikipedia.

6. Driven motion. Resonance. Important, but we'll not cover these here.

2 Background: The quantum harmonic oscillator

1. The Hamiltonian of the quantum harmonic oscillator:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2. \tag{7}$$

   Here, $\hat{p}$ and $\hat{x}$ are Hermitian operators satisfying the commutator relation $[\hat{x}, \hat{p}] \equiv \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar I$.

2. Typical representation: state (or wave function) ψ is a normalized $\mathbb{R} \to \mathbb{C}$ function, $\hat{x}\psi(x) = x\psi(x)$ and $\hat{p}\psi(x) = -i\hbar\partial_x\psi(x)$.

3. I'll omit the hats from now on: $\hat{H} \mapsto H$, $\hat{x} \mapsto x$, etc.

4. The energy spectrum of the quantum harmonic oscillator:

$$E_n = \hbar\omega(n + 1/2), \text{ with } n \in \{0, 1, 2, \dots\}. \tag{8}$$

5. The ground state wave function of the quantum harmonic oscillator:

$$\psi_0(x) = \frac{1}{\pi^{1/4}\sqrt{\ell}} e^{-\frac{x^2}{2\ell^2}}, \tag{9}$$

where $\ell = \sqrt{\frac{\hbar}{m\omega}}$ is the *oscillator length* or *ground-state size* of the oscillator.

6. For the excited states, see Figure 2, Wikipedia, textbooks, etc.

7. Notation for the oscillator energy eigenstates: $\psi_0 \mapsto |0\rangle$, $\psi_1 \mapsto |1\rangle$, ..., $\psi_n \mapsto |n\rangle$,...

8. Exercise: For a mass-and-spring system with $m = 1$ kg, $k = 1\,\mathrm{N/m}$, calculate the eigenfrequency ω, the energy distance $\hbar\omega$ between adjacent energy eigenvalues, and the oscillator length ℓ.

9. Exercise: Consider an electron that lives in one dimension and is trapped in a harmonic oscillator confinement potential with an oscillator length of $\ell = 100$ nm.
(a) Compute the energy gap $\hbar\omega$ of the spectrum and express it in electronvolts.
(b) Compute the force and the electric field strength that is needed to shift the center of the ground state by 100 nm.
(c) Compute the force and the electric field strength that is needed to shift the ground-state energy by 1 meV.

3 Ladder operators and the number operator

1. Here, we describe the quantum harmonic oscillator in an 'algebraic' way, without referring to differential equations. This is useful, as this algebraic approach simplifies many calculations.

2. Definition of the *ladder operators* of the quantum harmonic oscillator:

Lowering operator:

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(x + \frac{i}{m\omega} p \right) \equiv \frac{1}{\sqrt{2}} \left(\frac{x}{\ell} + i\ell \frac{p}{\hbar} \right) \tag{10}$$

Since both x and p are Hermitian operators, it holds that

$$a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(x - \frac{i}{m\omega} p \right) \tag{11}$$

The Hermitian conjugate $a^\dagger$ of the lowering operator a is called *the raising operator*. The lowering and raising operators together are called the *ladder operators*. They are dimensionless.

3. Note that $a^\dagger \neq a$, and hence a and $a^\dagger$ are NOT Hermitian, and hence are not observables.

4. The definitions of the lowering and raising operators can be 'inverted', yielding:

$$x = \frac{\ell}{\sqrt{2}}\left(a + a^\dagger\right), \tag{12}$$

$$p = \frac{i\hbar}{\sqrt{2}\ell}\left(a^\dagger - a\right). \tag{13}$$

5. Commutation relation of the ladder operators, following from $[x, p] = i\hbar I$:

$$[a, a^\dagger] = I. \tag{14}$$

6. Definition of the *number operator*: $N = a^\dagger a$. It is Hermitian.

7. Spectrum (that is, the set of eigenvalues) of the number operator: natural numbers ($\mathbb{N}$).

8. Relation between the harmonic oscillator Hamiltonian, the number operator, and the ladder operators:

$$H = \hbar\omega(N + 1/2) = \hbar\omega(a^\dagger a + 1/2). \tag{15}$$

9. Relation between the eigenvectors of H (and N) and the ladder operators:

$$a\,|n\rangle = \sqrt{n}\,|n-1\rangle, \tag{16}$$

$$a^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle. \tag{17}$$

10. Exercise:
(a) Express the matrix representation of a, $a^\dagger$, and the number operator $N = a^\dagger a$, in the (truncated) basis of the energy eigenstates $|0\rangle, |1\rangle, |2\rangle$.
(b) Denote the above 3×3 matrices as $\tilde{a}$, $\tilde{a}^\dagger$, and $\tilde{N}$. Is $\tilde{a}^\dagger\tilde{a}$ equal to $\tilde{N}$?
(c) Do $\tilde{a}$ and $\tilde{a}^\dagger$ satisfy the canonical commutator relations of the ladder operators? That is, does it hold that $[\tilde{a}, \tilde{a}^\dagger] = I$?

11. Exercise: Compute $\langle x\rangle$, $\langle p\rangle$, $\langle x^2\rangle$, and $\langle p^2\rangle$ in the ground state $|0\rangle$ of the harmonic oscillator of mass m and eigenfrequency ω. Show that the Heisenberg inequality (aka Heisenberg uncertainty relation) is fulfilled; in fact, it is saturated.

Solution:

$$\langle x\rangle_{|0\rangle} = \langle 0|x|0\rangle = \langle 0|\frac{\ell}{\sqrt{2}}(a + a^\dagger)|0\rangle$$

$$= \frac{\ell}{\sqrt{2}}\left(\langle 0|a|0\rangle + \langle 0|a^\dagger|0\rangle\right) = 0, \tag{18}$$

$$\langle p\rangle_{|0\rangle} = 0, \tag{19}$$

$$\langle x^2\rangle_{|0\rangle} = \langle 0|x^2|0\rangle = \frac{\ell^2}{2}\langle 0|aa + aa^\dagger + a^\dagger a + a^\dagger a^\dagger|0\rangle$$

$$= \frac{\ell^2}{2}\langle 0|aa^\dagger + a^\dagger a|0\rangle = \frac{\ell^2}{2}\langle 0|I + a^\dagger a + a^\dagger a|0\rangle = \frac{\ell^2}{2}, \tag{20}$$

$$\langle p^2\rangle_{|0\rangle} = \frac{\hbar^2}{2\ell^2}, \tag{21}$$

$$(\Delta x)_{|0\rangle}(\Delta p)_{|0\rangle} = \hbar/2. \tag{22}$$

Figure 1(b) visualizes the ground state as a 'cloud' on the x-p plane.

12. Exercise: Consider the state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Compute the expectation value of the position and the momentum in this state. Repeat for $|\chi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$.

13. Exercise: Given m and ω, compute the time evolution of the state $|\psi\rangle$ above, and from that, compute the time evolution of the expectation value of the position and of the momentum.

14. Exercise: Does the time evolution of the position and momentum in the previous exercise satisfy the equation $m\frac{d\langle x\rangle_{\psi(t)}}{dt} = \langle p\rangle_{\psi(t)}$?

15. Exercise: Compute the positional dependence of $\psi_1(x)$ by combining (i) the ground-state wave function $\psi_0(x)$ in Eq. (9), (ii) the relation $a^\dagger|0\rangle = |1\rangle$, and (iii) the form of $a^\dagger$ given in Eq. (11).

4 Coherent states of the quantum harmonic oscillator

1. Coherent states are the states that most closely resemble classical motion of the quantum harmonic oscillator. (Further motivation: get some practice in using ladder operators; relevance to the physics of lasers; relevance to quantum sensing and metrology.)

2. We can associate a 'coherent state' to any complex number $\alpha \in \mathbb{C}$.

3. Definition: for $\alpha \in \mathbb{C}$, the coherent state $|\alpha\rangle$ is defined as the eigenstate of a with eigenvalue α:

$$a\,|\alpha\rangle = \alpha\,|\alpha\rangle\,. \tag{23}$$

   (Recall that a is not Hermitian, hence it might have non-real eigenvalues.)

4. Solving the eigenvalue equation yields

$$|\alpha\rangle = e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,. \tag{24}$$

5. Observation: in a coherent state, the distribution of 'energy quanta' follows a Poisson distribution. If we measure the number of energy quanta (formally, the observable $N = a^\dagger a$), then we find n with the probability

$$p_n = \frac{|\alpha|^{2n}}{n!}e^{-|\alpha|^2}, \tag{25}$$

   which is indeed a Poisson distribution with parameter $\lambda = |\alpha|^2$.

6. Claim: The real part of α encodes the (dimensionless) expectation value of the position. The imaginary part of α encodes the (dimensionless) expectation value of the momentum.

Exercise: Compute the position and momentum expectation values of $|\alpha\rangle$.

Hint: For a (not necessarily Hermitian) linear operator M it holds that $\langle\varphi|M^\dagger|\psi\rangle = \langle\psi|M|\varphi\rangle^*$.

Solution:

$$\begin{aligned}\langle x\rangle_{|\alpha\rangle} &= \langle\alpha|x|\alpha\rangle = \langle\alpha|\frac{\ell}{\sqrt{2}}(a+a^\dagger)|\alpha\rangle = \frac{\ell}{\sqrt{2}}\left(\langle\alpha|a|\alpha\rangle + \langle\alpha|a^\dagger|\alpha\rangle\right) \\ &= \frac{\ell}{\sqrt{2}}\left(\langle\alpha|a|\alpha\rangle + \langle\alpha|a|\alpha\rangle^*\right) = \sqrt{2}\,\ell\,\mathrm{Re}(\alpha). \qquad (26)\\ \langle p\rangle_{|\alpha\rangle} &= \frac{\sqrt{2}\hbar}{\ell}\mathrm{Im}(\alpha). \qquad (27)\end{aligned}$$

7. Claim: any coherent state $|\alpha\rangle$ saturates the Heisenberg uncertainty relation.

   Exercise: Compute the position and momentum standard deviations Δx and Δp for $|\alpha\rangle$.

   Solution:

$$\begin{aligned}(\Delta x)_{|\alpha\rangle} &= \sqrt{\langle x^2\rangle_{|\alpha\rangle} - \langle x\rangle^2_{|\alpha\rangle}} = \frac{\ell}{\sqrt{2}}, \text{ since} \qquad (28)\\ \langle x^2\rangle_{|\alpha\rangle} &= \frac{\ell^2}{2}\left(\alpha^2 + (\alpha^*)^2 + 2|\alpha|^2 + 1\right), \text{ and} \qquad (29)\\ \langle x\rangle^2_{|\alpha\rangle} &= 2\ell^2\left[\mathrm{Re}(\alpha)\right]^2 = \frac{\ell^2}{2}\left(\alpha^2 + (\alpha^*)^2 + 2|\alpha|^2\right). \qquad (30)\\ (\Delta p)_{|\alpha\rangle} &= \frac{\hbar}{\sqrt{2}\ell}. \qquad (31)\end{aligned}$$

   Eqs. (28) and (31) imply that the coherent state $|\alpha\rangle$ saturates the Heisenberg uncertainty relation.

   Figure 1(c) visualizes a coherent state as a 'cloud' on the x-p plane – same size as the ground state, but displaced.

8. Time evolution starting from a coherent state $|\alpha_0\rangle$, subject to the harmonic oscillator Hamiltonian:

$$|\psi(t)\rangle = e^{-i\omega t/2}\,|e^{-i\omega t}\alpha_0\rangle \qquad (32)$$

   Derivation is straightforward, starting from Eq. (24). Note that 'global phase factor' $e^{-i\omega t/2}$ is irrelevant.

   The result (32) implies the dynamics shown in Figure 1(d).

9. Exercise: Express the time evolution of the position and momentum expectation values of the time-evolving coherent state given in Eq. (32).

10. Exercise: Does the raising operator $a^\dagger$ have any eigenvectors?

Appendix

Recommended videos: The youtube series 'Professor M Does Science' has quite a few related useful videos, for example:

1. The quantum harmonic oscillator
   `https://www.youtube.com/watch?v=0dizRUe84bg`
   11-min video introduction to the topic, mostly reviewing the classical harmonic oscillator.

2. Ladder and number operators of the quantum harmonic oscillator
   `https://www.youtube.com/watch?v=Kb9twGd25P0`
   15-min video. Overlaps with my section on this topic (see above), but the video also contains proofs and derivations.

3. Coherent states in quantum mechanics
   `https://www.youtube.com/watch?v=x0wk98uMyys`
   22-min video containing derivations, proofs related to the coherent states.